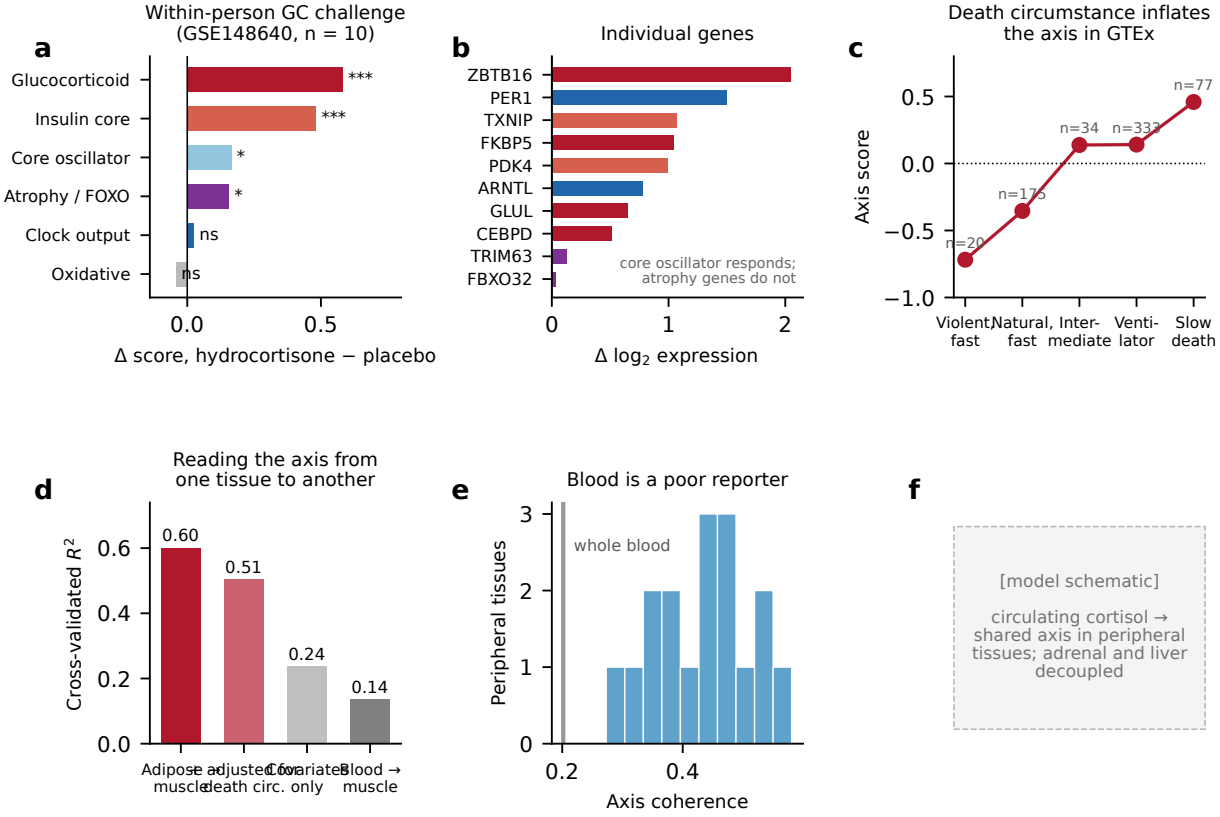
a
Within-person GC challenge
(GSE148640, n = 10)
Glucocorticoid
Insulin core
Core oscillator
Atrophy / FOXO
Clock output
Oxidative
***
***
*
*
ns
ns
0.0
0.5
Δ score, hydrocortisone − placebo
b
Individual genes
ZBTB16
PER1
TXNIP
FKBP5
PDK4
ARNTL
GLUL
CEBPD
TRIM63
FBXO32
core oscillator responds;
atrophy genes do not
0
1
2
Δ log2 expression
c
Death circumstance inflates
the axis in GTEx
Axis score
0.5
0.0
−0.5
−1.0
n=20
n=175
n=34
n=332
n=77
Violent,
fast
Natural,
fast
Inter-
mediate
Venti-
lator
Slow
death
d
Reading the axis from
one tissue to another
Cross-validated R2
0.0
0.2
0.4
0.6
0.60
0.51
0.24
0.14
Adipose →
muscle
+ adjusted
death circ.
Covariates
only
Blood →
muscle
e
Blood is a poor reporter
Peripheral tissues
whole blood
0
1
2
3
0.2
0.4
Axis coherence
f
[model schematic]
circulating cortisol →
shared axis in peripheral
tissues; adrenal and liver
decoupled